

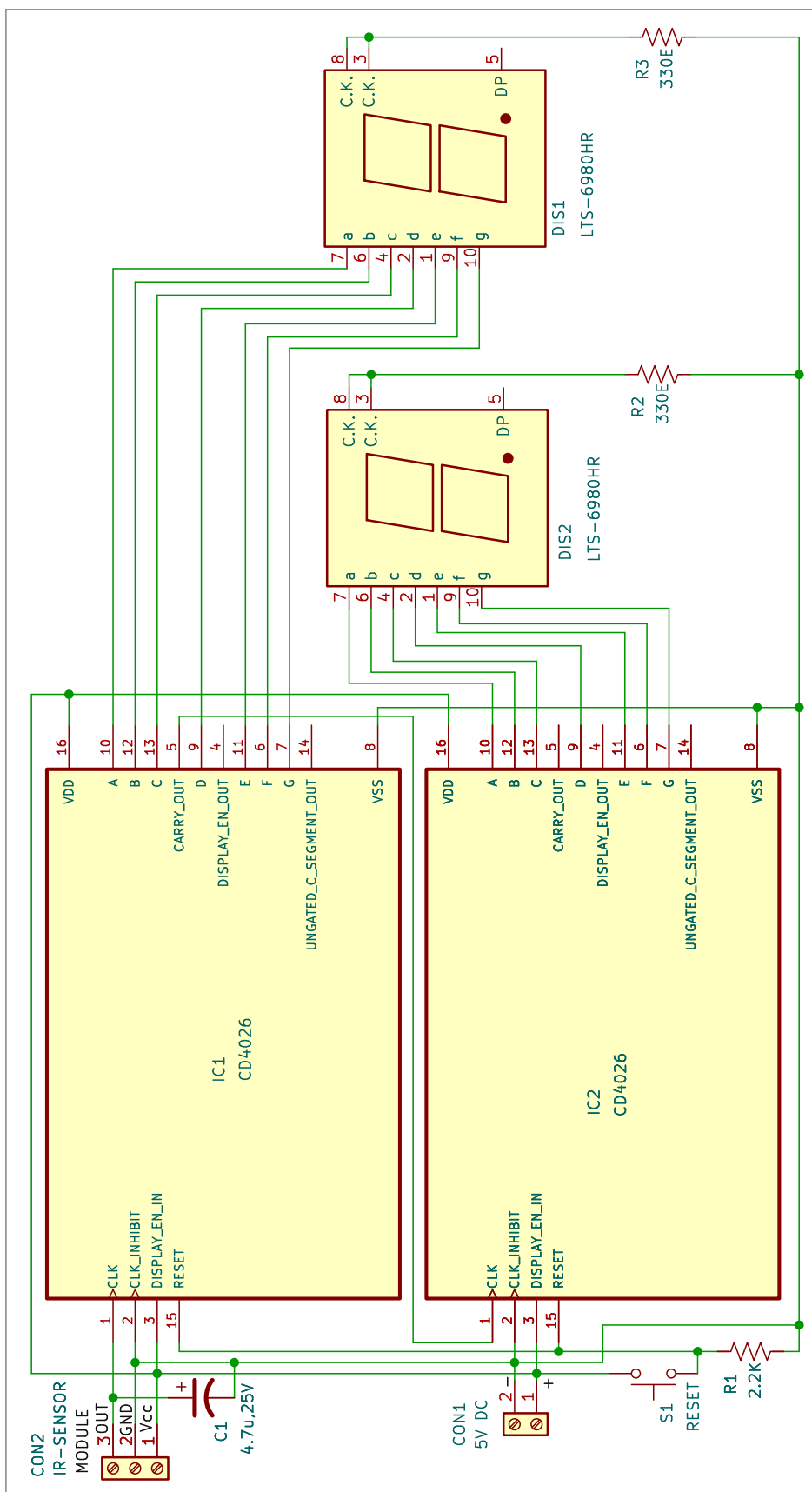


Fig. 2: Circuit diagram

input of the first counter IC (IC1), while the carry-out output of IC1 is connected to the clock input of IC2 for cascading operation. The outputs of both ICs are wired to their respective 7-segment displays using the correct pin configuration.

For testing, apply a regulated 5V DC supply to the circuit. Initially, both displays should show 00, indicating the reset condition. Move a hand or walk in front of the IR sensor to simulate entry detection. The count should increase by one for each detection. Continue testing to verify correct counting up to 99 and ensure that the tens digit increments correctly after every ten counts. Press the reset switch to confirm that the display returns to zero instantly. If the counter skips counts or behaves erratically, check the IR sensor alignment, verify all wiring connections, and ensure that the 5V power supply is stable and free from electrical noise.

Installation

Installation of this counter requires proper placement and mounting to ensure accurate operation. The system should be installed at the only entry or exit point, such as a main door or corridor, so that every person is counted without fail.

Mount the IR sensor at a height of approximately 90 to 120 centimetres on one side of the doorway, aligned with the expected path of movement. Ensure that the sensor is not exposed to direct sunlight or highly reflective surfaces, as these may cause false triggering. The display unit should be installed in a clearly visible